ROHIT PACHORI

Senior Software Engineer | Kotlin • Jetpack Compose • MVVM • MVI • Clean Architecture | Flutter | Dart | Bloc
Bengaluru, India | +91-8277270451 | rpachori21@gmail.com | github.com/githubrohitgithub | linkedin.com/in/rohit-pachori | githubrohitgithub.github.io/portfolio

SUMMARY

Senior mobile engineer with ~8 years of experience in native Android (Kotlin, Jetpack Compose) and 5+ years in Flutter (Dart). Owns core native modules of a self-service kiosk platform live across 800+ Compass Group sites in 14+ countries. Built TDS and IMD from scratch in Flutter, live in 14+ countries, and own both end to end: production issues, releases and feature development. Strengths: offline-first and real-time architecture, payment and hardware integrations (printers, scanners), and production stability: debugging, memory and performance tuning for kiosks that run all day. Also ships privacy-first, on-device AI products (RAG with local LLMs, ML Kit, LiteRT, Gemini Nano).

CORE SKILLS

Android: Kotlin, Jetpack Compose, Coroutines & Flow, Hilt, Room, WorkManager, Navigation, Paging, CameraX, ExoPlayer, multi-module architecture, Flutter, Dart, Bloc, Provider, Debugging, Unit Testing, Integration Testing
Architecture: Clean Architecture, MVI, MVVM, Repository pattern
Stability & Quality: Crashlytics, ANR analysis, memory profiling, ProGuard/R8, JUnit, Mockito, Espresso
Data & Backend: Firebase (Realtime DB, Firestore, FCM, Remote Config, Auth), Retrofit, REST APIs, SQLite
Hardware & Payments: ESC/POS printers (Sunmi, Zebra, Bixolon), barcode/QR scanners, BLE/IoT, FreedomPay, Pine Labs, Razorpay
On-device AI: ML Kit (OCR, Object Detection), Google AI Edge LiteRT (TFLite), Gemini Nano, Ollama, RAG (FAISS), Whisper.cpp, Piper TTS
AI-Assisted Engineering: Cursor, Claude, Gemini; agent-assisted debugging, test generation and code review
Also: Flutter (5+ years), Dart, BLoC, Provider, Hive, Java, Git, Gradle, CI/CD, Play Console

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Native Android | May 2022 - Present

SmartQ (Bottle Lab Technology Pvt Ltd), Bengaluru, technology arm of Compass Group

- Own production support for kiosk apps running across 800+ sites: triage through Crashlytics and logs, root-cause field issues and ship hotfixes.
- Resolved out-of-memory crashes and unnecessary Compose recompositions on low-end kiosk devices loading large menus; root-caused the problem to memory pressure at menu scale and re-architected the entire menu feature and module (repository-mapped read models, use cases, single reducer to UI state) so cart actions no longer reload or recompose the menu list.
- Fixed out-of-memory crashes in a Flutter app when large volumes of orders arrived on low-end devices, so it stays stable under peak load in high-traffic canteens.
- Architected the real-time order-status and pending-count system as a single shared Firebase listener, replacing per-screen listeners; cutting redundant data transfer while keeping all screens in sync.
- Led the migration of 6+ core modules to Jetpack Compose with Clean Architecture and MVI, now the team standard for new native modules.
- Designed the meal-deal cart data model spanning 3 modules, resolving structural and data-integrity issues before implementation; adopted as the team's reference architecture for the feature.
- Built offline-first order handling with WorkManager and integrated FreedomPay and Pine Labs payments (card, contactless, QR) plus Sunmi/Zebra/Bixolon printers and scanners via ESC/POS.
- Implemented store-closure automation and multi-region tax handling (including German TSE/Fiskaly-compliant receipts) to meet compliance requirements across countries.
- Built TDS and IMD from scratch in Flutter, live in 14+ countries, and own both end to end (production issues, releases, feature development); led a native-to-Flutter migration of select modules, reducing maintenance overhead by ~40%; kept performance-critical and hardware-integrated modules in native Kotlin/Compose.

Software Engineer, Native Android | Aug 2021 - Apr 2022

Cognitive Cloud Pvt Ltd, Bengaluru

- Built native Kotlin healthcare apps handling patient data with local encryption and concurrent threading for real-time vitals processing.
- Introduced TDD with JUnit and Mockito, achieving a 25% reduction in post-launch crash rates; performed code reviews on sensitive-data handling.

Software Engineer, Native Android | Jan 2019 - Jul 2021

Aislyn Technologies Pvt Ltd, Bengaluru

- Built Aislyn Learn (Kotlin): 15+ paid course tracks with Razorpay checkout, Google Sign-In and ExoPlayer video streaming.
- Built QR Smart Parking (Kotlin): ZXing QR scanning, live slot tracking, prepaid wallet and Retrofit-based REST integration.
- Developed Android apps interfacing with BLE peripherals and IoT health sensors; built Dementia Care Companion in Flutter, polling remote vitals (SpO2, BP) from Realtime Database.

FEATURED PROJECTS

Food Ordering Kiosk, Campus Dining Platform (Native Android) | 2024 - Present

- Self-service Kotlin/Compose kiosk for 50+ restaurant vendors with receipt printers, barcode/ID scanners and card terminals; meal-deal combo pricing, ID-based subsidies and multi-provider payments.
  Stack: Kotlin, Jetpack Compose, Firebase, Room, Hilt, WorkManager

TDS (Token Display System) & IMD (Intelligent Menu Display), Flutter | 2023 - Present

- Built both from scratch and own them end to end (production support, releases, feature development), live in 14+ countries. TDS: Android TV order display with real-time status and a cloud-managed banner/media system (video, image, Lottie), using BLoC and Firebase Remote Config for country-specific themes without app updates.
- IMD: multi-restaurant menu carousel syncing live pricing and availability via Firebase Realtime Database, with Hive offline fallback and admin feature flags.

SecureVault, Encrypted Password Manager (Flutter) | Open source

- Local-only credential manager with AES-encrypted storage and biometric authentication, zero cloud exposure of sensitive data; builds for Android, iOS and Web.

AI Object Detection, Native Android | 2026

- Jetpack Compose app doing real-time on-device object detection with ML Kit and Google AI Edge LiteRT (SSD MobileNet) over CameraX, with live switching between detection pipelines.
- Added IoU-based temporal stabilization to stop bounding-box flicker, plus touch-to-highlight and pinch-to-zoom controls.
  Stack: Kotlin, Jetpack Compose, LiteRT (TFLite), ML Kit, CameraX

AI INTEGRATION

Privacy-first AI products I designed and shipped end to end. All three run fully on-device with no cloud APIs.

RoPac, Offline Personal AI Assistant (Flutter + Python) | 2026

- Flutter macOS app talking to a Python bridge that runs RAG over the user's documents (PDF, Excel, Word, code) and a local Ollama LLM; memory, knowledge and voice models live in one portable folder.
- Built the retrieval pipeline: chunking, embeddings, hybrid search and a FAISS index, with cross-chunk conflict detection that flags contradicting passages in the prompt.
- Added offline voice mode (Whisper.cpp speech recognition, Piper TTS) and PBKDF2 owner-password protection for long-term memory.
  Stack: Flutter, Python, Ollama, FAISS, Whisper.cpp, Piper, PBKDF2

MediLens, AI Healthcare Document Assistant (Native Android) | github.com/githubrohitgithub/Medi-Lense

- Offline-first Kotlin/Compose app: CameraX capture and ML Kit OCR read prescriptions and lab reports, and a local parser extracts medicine names, doses, frequencies and dates.
- Room-backed offline search across title, summary, text and medicines; plain-language on-device summary, with an optional Gemini Nano rewrite via ML Kit GenAI when the model is on the phone.
- Daily medication reminders with WorkManager; images and database kept app-private with Android backup turned off.
  Stack: Kotlin, Jetpack Compose, MVVM, CameraX, ML Kit OCR, Gemini Nano, Room, WorkManager

VisionAssist, AI Accessibility Assistant (Native Android) | github.com/githubrohitgithub/VisionAssist

- Real-time camera assistant for visually impaired users: CameraX streams frames to ML Kit Object Detection and OCR, and a scene interpreter turns bounding boxes into spoken left / ahead / right descriptions via Android TextToSpeech.
- Caution message and haptic feedback when a large object fills the center of the frame; OCR throttled to a 1.5s interval to keep the camera preview smooth.
- TalkBack-aware speech to avoid double announcements; high-contrast dark UI with 56dp controls and live-region status.
  Stack: Kotlin, Jetpack Compose, MVVM, CameraX, ML Kit Object Detection, ML Kit OCR, TextToSpeech

ADDITIONAL PROJECTS

- **AislynTech Engineering Projects Platform (Flutter, Android and iOS):** marketplace with 19 project categories, cart, wishlist and Razorpay checkout, plus student and employee portals. Google Sign-In via Firebase Auth, MVVM + Repository with Provider, 5-minute response cache, PDF viewer, REST APIs across two backends.
- **Dog Health, IoT Pet Neckband App (Flutter), 2021:** live heart rate and temperature from an IoT neckband API, GPS tracking with Google Maps, vet directory and reports. Firebase Auth, Google Sign-In, SQLite local storage.
- **Face Recognition (Flutter, ML Kit), 2023:** register faces from camera or gallery, real-time detection and recognition with Google ML Kit, offline matching against a Hive local face database, spoken feedback via TTS.

AWARDS & RECOGNITION

- Hackathon Winner, Compass Group UK & I (awarded by Simon Hill, Digital Director), 2024
- Above and Beyond, Emerging Star, SmartQ, Bengaluru, 2024

CERTIFICATIONS & WORKSHOPS

- Core Java & J2EE, Glovish Technologies, Bengaluru
- Java and Testing, ABC for Java and Testing, Bengaluru
- Participation: Ethical Hacking (Vasavi Jnana Peetha College), AI & NLP, FlutterFlow Workshop, Bengaluru

EDUCATION

- Bachelor of Computer Applications (BCA), East West College, Bangalore University | CGPA 7.5
- Higher Secondary (12th, Science) | 75% and High School (10th, Science) | 62%, Hema Convent School, MP Board